Question	Correct Answer	Full Justification of the Correct Answer
1	B	On-demand self-service means users can provision, change, or release computing capabilities (like server time or network storage) automatically and immediately without requiring human interaction with the service provider.
2	C	IaaS (Infrastructure as a Service) provides the core computing infrastructure (VMs, storage, networks). The consumer has control over the operating system, storage, deployed applications, and network components.
3	B	The Hypervisor is the essential software or firmware layer that creates the virtualization environment, manages the physical hardware resources, and allocates them to the guest operating systems running inside the VMs.
4	B	A Private Cloud is infrastructure provisioned for exclusive use by a single organization. It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.
5	C	Containers share the host OS kernel and virtualize the OS layer. VMs run their own isolated guest OS, which virtualizes the hardware layer via the hypervisor.
6	C	The Dockerfile is a text file containing all the commands, in order, needed to build a given Docker image.
7	D	A Pod is the smallest and simplest unit in the Kubernetes object model. It represents a single instance of a running process in a cluster, usually wrapping a single application container.
8	D	Microsoft 365 (online productivity suite) is a prime example of SaaS because the user accesses the completed application via a browser without managing any underlying infrastructure, operating system, or application code.
9	C	A Type 1 Hypervisor (e.g., VMware ESXi, Microsoft Hyper-V) runs directly on the host hardware (bare metal) to manage guest operating systems, offering better performance and scalability than Type 2.
10	D	Over-subscription/Over-commitment (or sometimes density issues) refers to the act of allocating more virtual resources (like vCPUs or RAM) to VMs than the physical hardware actually possesses, relying on the fact that not all VMs will use their full allocation simultaneously.
11	C	Rapid Elasticity is the characteristic that allows resources to be quickly and elastically provisioned and released, scaling outward or inward automatically to meet demand.
12	B	Virtualization allows multiple logical entities (VMs) to run on one physical server, drastically increasing resource utilization and reducing the need for numerous physical machines (server consolidation).
13	C	The Hypervisor's core function is to ensure isolation between VMs and manage their controlled access to shared physical

		components like the CPU, memory, and I/O devices.
14	C	IaaS delivers fundamental computing resources, primarily Virtual Machines, virtual networks, and virtual storage, giving the user the most fundamental control over the infrastructure.
15	C	In PaaS, the cloud vendor manages the OS, runtime, and infrastructure. The consumer is responsible only for deploying their application code and managing the application's data.
16	C	Virtual Machine Provisioning (or instantiation) is the automated process that creates a new VM from an image, allocates resources, and performs post-creation configuration like IP assignment and user initialization.
17	C	RAID is a foundational technology for StaaS/cloud storage, as its primary role is to combine multiple physical drives into one or more logical units for the purpose of data redundancy and fault tolerance against disk failure.
18	B	The purpose of an HA (High-Availability) Cluster is to automatically detect failures (hardware or software) on a node and seamlessly move the workload (VM, application, or service) to a surviving node, ensuring business continuity.
19	B	OpenStack Horizon is the official web-based dashboard that provides a user-friendly interface for interacting with and managing various OpenStack services and cloud resources.
20	B	In OpenStack, a Flavor is a predefined template that defines the sizing (number of vCPUs, amount of RAM, and root disk space) for an instance (VM).
21	C	The syntax -p HostPort:ContainerPort maps the specified host machine port (8088) to the container's internal listening port (80), allowing external network access to the application.
22	B	Container orchestration systems like Kubernetes automate the complex task of deploying, scaling, and managing many individual containerized applications across a large cluster of machines, ensuring high availability.
23	C	A ReplicaSet ensures that a stable set of replica Pods are running at any given time. If a Pod fails, the ReplicaSet automatically creates a replacement.
24	C	Measured Service means resource usage is monitored, controlled, and reported, allowing providers to bill the consumer based on quantifiable usage metrics like CPU-hours, GB-transferred, or I/O operations.
25	B	Live Migration (or vMotion/Live Migration) is the capability to move a running VM from one physical host to another without interrupting the application or network service, essential for hardware maintenance and load balancing.
26	B	A Hybrid Cloud allows an organization to use a secure Private Cloud for sensitive data/regulatory needs while leveraging the Public Cloud's massive scale and elasticity for less sensitive, burstable workloads.
27	C	By outsourcing the management of the OS, Runtime, and Middleware, and focusing only on deploying their code, the company is primarily utilizing PaaS (Platform as a Service).

28	C	A Service in Kubernetes is an abstract way to expose an application running on a set of Pods as a network service, providing a stable IP address and DNS name regardless of Pod failure or scaling.
29	C	OS Virtualization (Containers) abstracts the operating system layer; it uses the host's existing kernel and isolates user-space processes (via technologies like cgroups and namespaces).
30	B	Docker Compose is a tool for defining and running multi-container Docker applications. It uses a YAML file to configure the application's services, networks, and volumes in one place.
31	B	Network Traffic Egress (data transferred out of the cloud region) is the most common resource that incurs significant charges across major public cloud vendors, while Ingress (data transferred in) is often free or minimally charged.
32	B	The correct high-level provisioning workflow is: 1. Select Flavor (sizing/hardware). 2. Select Image (OS/content). 3. Allocate Resources (Compute/Storage, ensuring capacity). 4. Network/IP Assignment (connecting the instance).
33	B	Unlike Type 2, a Type 1 Hypervisor acts as the first layer of software above the hardware, meaning the hypervisor itself must contain or manage the necessary device drivers for all underlying physical hardware, which is complex to maintain.
34	B	The Deployment object is the core controller for rolling updates. It describes the desired state and manages the lifecycle of the application by creating and managing the underlying ReplicaSets to achieve that state.
35	C	IaaS gives the consumer control down to the operating system level, allowing them to install custom libraries, modify kernel settings, and perform deep configuration necessary for highly specialized applications.
36	B	VM Sprawl occurs when a proliferation of unused, forgotten, or undocumented VMs are running (or stored), leading to wasted storage, unnecessary licensing fees, and a larger security attack surface.
37	B	Major public clouds invest heavily in a robust, high-speed, and redundant global network backbone (regions and availability zones) to ensure low-latency connectivity and massive scale worldwide.
38	B	The CMD instruction in a Dockerfile defines the default command that will be executed when the container starts. It is meant to be overridden by the user when running the container.
39	B	Utility Computing is a form of pricing where the consumer pays based on usage, similar to a utility service (like electricity or water), meaning they are charged only for the exact resources consumed, often measured in fine granular increments.
40	C	A Security Group acts as a stateful virtual firewall that applies to one or more virtual machine instances at the network interface level, controlling all traffic going to and coming from those instances.

> 1:

• Introduction to cloud
• What computing paradigms are there?
• Characteristics and benefits
• Understanding Cloud Vendors (AWS/Azure/GCP)
• Definition
• Characteristics
• Components

No Lab

> 2:

• Introduction to Virtualization
• Types of Hypervisors
• Cluster Architecture
• Cluster Requirements
• Virtual machine provisioning
• Virtualization applications in enterprises

• Pitfalls of virtualization
• Need of Virtualization Provisioning
• Work flow of Virtualization Provisioning
• Operating System Virtualization
• Challenges in Virtualization

Lab:

• VM Creation and Management
• Flavor, Images repository, instances
• Create Ubuntu22.04 based VM
• Access the VM through console
• Check the IP, Storage and Linux user account details

> 3:

• Introduction to SAAS
• Pros and Cons of SAAS Model
• Traditional packaged software Vs SAAS
• SAAS examples
• Introduction to IAAS
• Introduction to virtualization
• Types and Uses of Virtualization
• Virtual Machine Provisioning
• Virtual Machine Migration Services
• Private Cloud Computing Deployment
• Introduction to PaaS

• Storage as Service (RAID)
• Challenges of cloud environment
• Hypervisor
• Comparisons of web services
• Organizational Scenarios of Clouds

> 4:

• Administering & Monitoring cloud services
• Benefits and limitations
• Deploy application over cloud.
• Comparison among SAAS, PAAS, IAAS
• Cloud Computing Basics
• Cloud Products and Solutions
• Cloud Pricing
• Compute Products and Services

Lab:

• Explore Openstack Horizon Dashboard Interface
• Flavor, Images repository, instances
• VM Creation and Management
• Create Ubuntu22.04 based VM
• Access the VM through console
• Check the IP, Storage and Linux user account details
• Create and manage security groups
• SaaS, StaaS & PaaS service enablement on Cloud
• Create your own file hosting service just like Dropbox, using opensource platform ownCloud – SaaS

> 5:

• Concept of a container
• Introduction to Docker
• Installing Docker
• Using Docker to pull an image from Docker Hub
• Creating Docker file for Python, Java, and Web applications
• Docker commands – Image and Container-related
• Docker compose
• Interacting with a container

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• Container orchestration
• Introduction to Kubernetes
• Cluster, Node, Pod, Service, Replica set, Deployment
• Kubernetes installation
• Creating a pod
• Using yaml file for Kubernetes deployment
• Understanding replica sets and deployment using an example

• Deploying a container on the cloud
• Running the container orchestration on the cloud

Lab:
• Docker Installation and Configuration
• Docker Image Management
• Docker Port mapping
• Docker Container Management
• Install webserver server-specific container and access the web page
• Nginx/Apache